Separate untimed diagnosis of contraction and recursion

Root edges remaining after contraction

Total recursive KKT edge-input volume

Bipartite · 16-vertex side · 4,096
Bipartite · 16-vertex side · 16,384
Bipartite · 16-vertex side · 65,536
Bipartite · 16-vertex side · 262,144
Bipartite · 256-vertex side · 4,096
Bipartite · 256-vertex side · 16,384
Bipartite · 256-vertex side · 65,536
Bipartite · 256-vertex side · 262,144
Bipartite · floor(√n)-vertex side · 4,096
Bipartite · floor(√n)-vertex side · 16,384
Bipartite · floor(√n)-vertex side · 65,536
Bipartite · floor(√n)-vertex side · 262,144
Core with 50% attached leaves · 4,096
Core with 50% attached leaves · 16,384
Core with 50% attached leaves · 65,536
Core with 50% attached leaves · 262,144
Core with ≥90% attached leaves · 4,096
Core with ≥90% attached leaves · 16,384
Core with ≥90% attached leaves · 65,536
Core with ≥90% attached leaves · 262,144
Ring of 16-vertex cliques · 4,096
Ring of 16-vertex cliques · 16,384
Ring of 16-vertex cliques · 65,536
Ring of 16-vertex cliques · 262,144
2D grid · random weights · 4,096
2D grid · random weights · 16,384
2D grid · random weights · 65,536
2D grid · random weights · 262,144
2D grid · distance weights · 4,096
2D grid · distance weights · 16,384
2D grid · distance weights · 65,536
2D grid · distance weights · 262,144
3D grid · random weights · 4,096
3D grid · random weights · 16,384
3D grid · random weights · 65,536
3D grid · random weights · 262,144
3D grid · distance weights · 4,096
3D grid · distance weights · 16,384
3D grid · distance weights · 65,536
3D grid · distance weights · 262,144
8-regular graph · 4,096
8-regular graph · 16,384
8-regular graph · 65,536
8-regular graph · 262,144
Hub control · 4,096
Hub control · 16,384
Hub control · 65,536
Hub control · 262,144
Random m=4n control · 4,096
Random m=4n control · 16,384
Random m=4n control · 65,536
Random m=4n control · 262,144

0%
25%
50%
75%
100%
Remaining root edges / original m

1.0
1.1
1.2
1.3
1.4
1.5
Σ subproblem edges / original m (root included)

Arithmetic seed means; bars span graph-level means. Zero remaining edges includes an earlier one-round finish.
Edge-input volume excludes repeated internal passes. These counters do not isolate a causal sampling speedup.